

Autonomous AI-Powered SOC Analytics System

Integrated Threat Detection & Intelligence Pipeline

Presented by

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Course

AI in Cybersecurity based on NVIDIA Morpheu

Problem Statement

The Challenge

SOC analysts are overwhelmed by log volume and alert fatigue

Manual Limitation

.Human triage cannot keep up with high-frequency network anomalies

Our Solution

.An autonomous pipeline that automates the "Detection-to-Analysis" lifecycle

System Architecture

Design Philosophy

.Modular integration of ML-based detection and Generative AI Agents

Key Components

.Data Pipeline: Simulated Kafka log ingestion

Orchestration: Microsoft Agent Framework managing Tier
.stnega trepxE ITC &1

Intelligence: Llama-3.3-70b (via Groq API) for rapid threat interpretation.

Results & MITRE ATT&CK Mapping

The "Integrated" Value

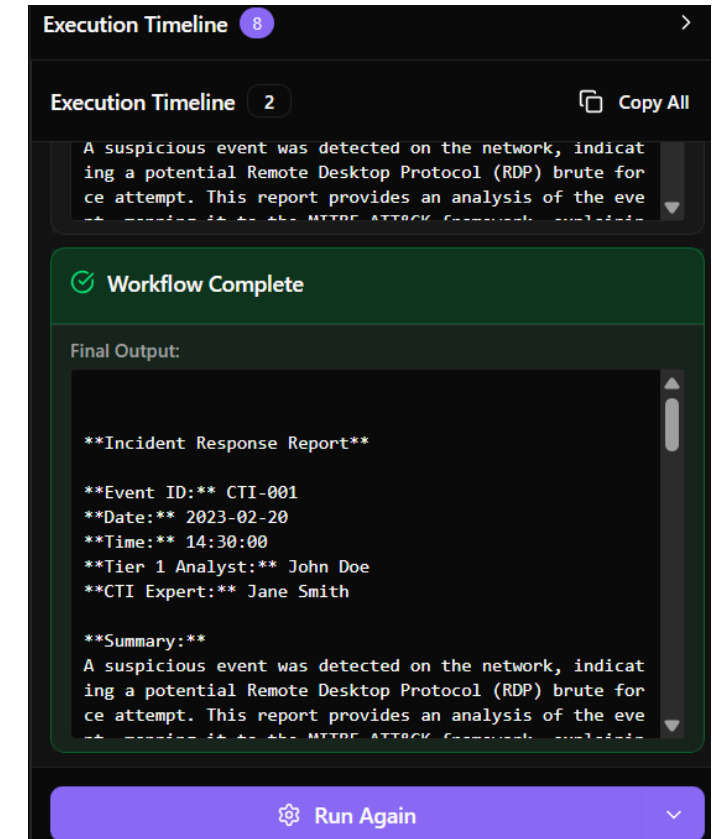
.Automating threat classification using universal frameworks

Mapping

Every identified event is mapped to MITRE ATT&CK Tactic & Technique
(e.g., T.)ecroF eturB -1110

Key Achievement

.Transformed raw log data into actionable Incident Response Reports automatically



Live Demonstration

Video Demo

.Full end-to-end workflow execution

Watch here

<https://youtu.be/20HiVw677Bo>

Conclusions

Integration Success

Successfully combined ML detection with Multi-Agent AI
.systems

Operational Impact

.Demonstrated significant reduction in manual analysis time

Future Scope

.Scalability to enterprise SIEM platforms (e.g., Splunk integration)